CTA200 Assignment 3

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1 The Mandelbrot Set

1.1 diverge_when(c, z0=0, max_iter=30)

Finds whether a given point c on the complex plane converges or diverges after iterating through the sequence $z_{i+1} = z_i^2 + c$.

Parameters:

- c: *complex*
  The point on the complex plane.
- z0: *float, optional*
  The first term of the sequence.
- max_iter: *int, optional*
  The maximum number of terms to calculate before a verdict is reached.

Returns:

- n: *int*
  The number of iterations it took for the sequence to diverge. Divergence defined as $||z_i|| > 100$. If the point does not diverge, then returns None.

1.2 combine(real, imag)

Takes the real part and the imaginary part of a complex number and combines them into the whole complex number.

Parameters:

- real: *array-like*
  The real part of the complex number.
- imag: *array-like*
  The imaginary part of the complex number.

Returns:

- c_num: *array-like*
  The whole complex number.

1.3 Plots and result

This is the well-known Mandelbrot set. Figure 1 displays the set coloured two different ways, one for whether each point diverges and the other for how fast each point diverges.

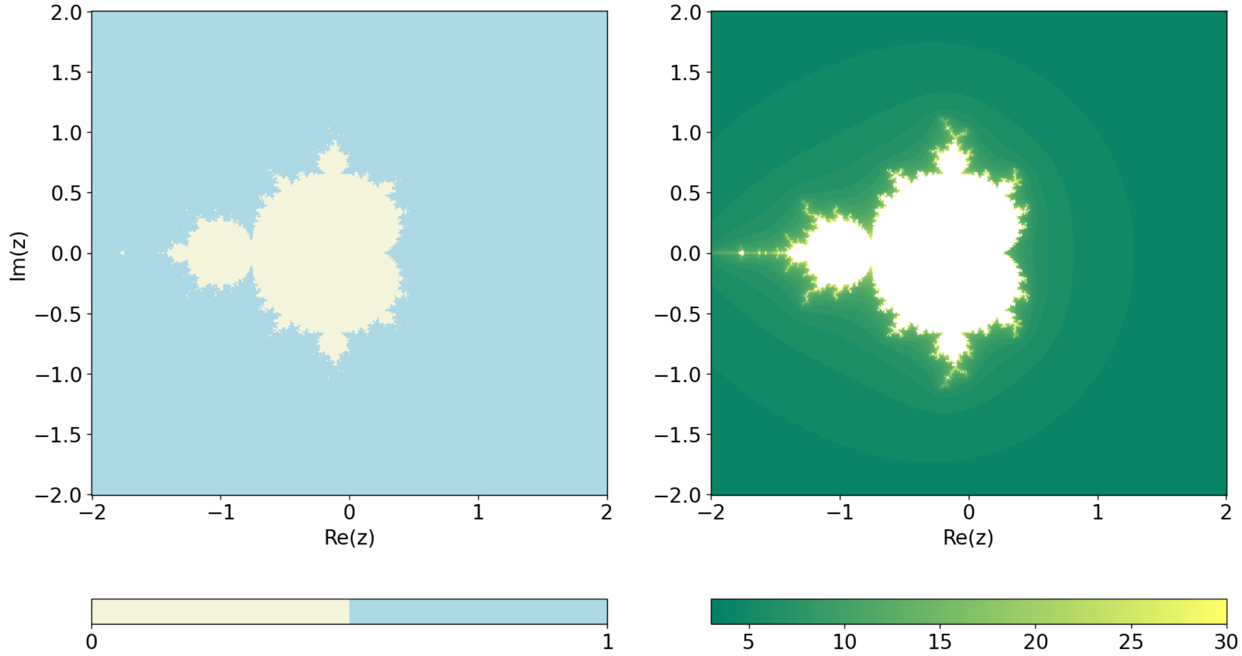

Figure 1: The Mandelbrot set, coloured two different ways. Left: coloured according to whether each point diverges or is bounded. On the colourbar, 0 corresponds to bounded and 1 corresponds to diverging. Right: coloured according to how fast each point diverges. White corresponds to no divergence, and the colourbar denotes how many terms of the sequence it took to diverge.

2 Lorenz Systems

2.1 lorenz(t, v, sigma, r, b)

Vectorized system of Lorenz equations used for the right-hand side of solve_ivp. Uses the following three first-order differential equations:

$$\dot{X} = -\sigma(X - Y) \tag{1}$$

$$\dot{Y} = rX - Y - XZ \tag{2}$$

$$\dot{Z} = -bZ + XY \tag{3}$$

Parameters:

- t: *array-like*
  Time.
- v: *array-like*
  The vectorized coordinates [X, Y, Z].
- sigma: *float*
  The Prandtl number, defined as the ratio of the kinematic viscosity to the thermal diffusivity.
- r: *float*
  The Rayleigh number.
- b: *float*
  The length scale.

Returns:

- vdot: *array-like*
  The vectorized time derivatives $[\dot{X}, \dot{Y}, \dot{Z}]$.

2.2 distance(r, r_p)

Find the distance between two points r and r_p.

Parameters:

- r: *array-like*
  The first coordinate point.

- r_p: *array-like*
  The second coordinate point.

Returns:

- d: *float*
  The distance between the two points.

2.3 Plots and result

Figures 2 and 3 are reproductions from Lorenz' paper. I hypothesize that their slight incongruence to Lorenz' figures are likely due to a difference in numerical solver. Lorenz' plotted solution to the convection equations seems to evolve for longer than ours, which may mean that his timestep was greater.

Figure 4 displays the distance between two solutions with only slightly differing initial conditions. We see on the semilog plot that a difference of 1e-8 in only one coordinate causes the distance between solutions to grow exponentially, leaving the end states of each solution very far apart in phase space.

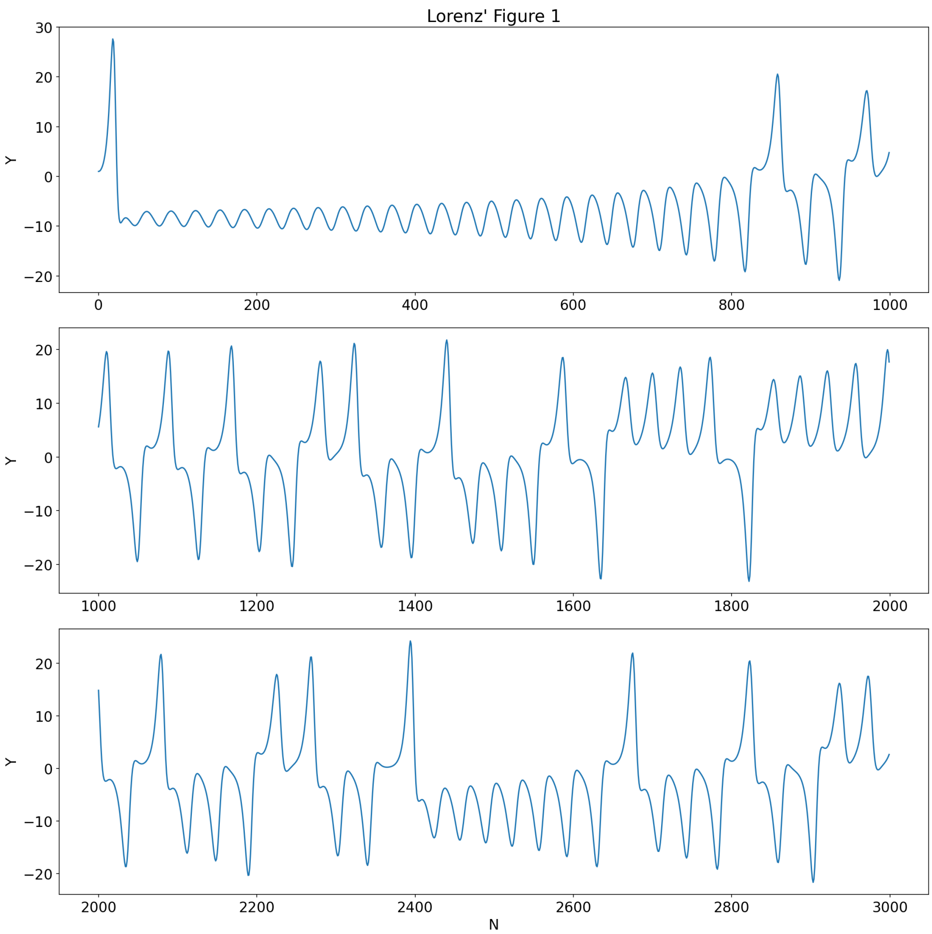

Figure 2: Reproduced Figure 1 from Lorenz' paper. Numerical solution of the convection equations. First 60 units of time are shown.

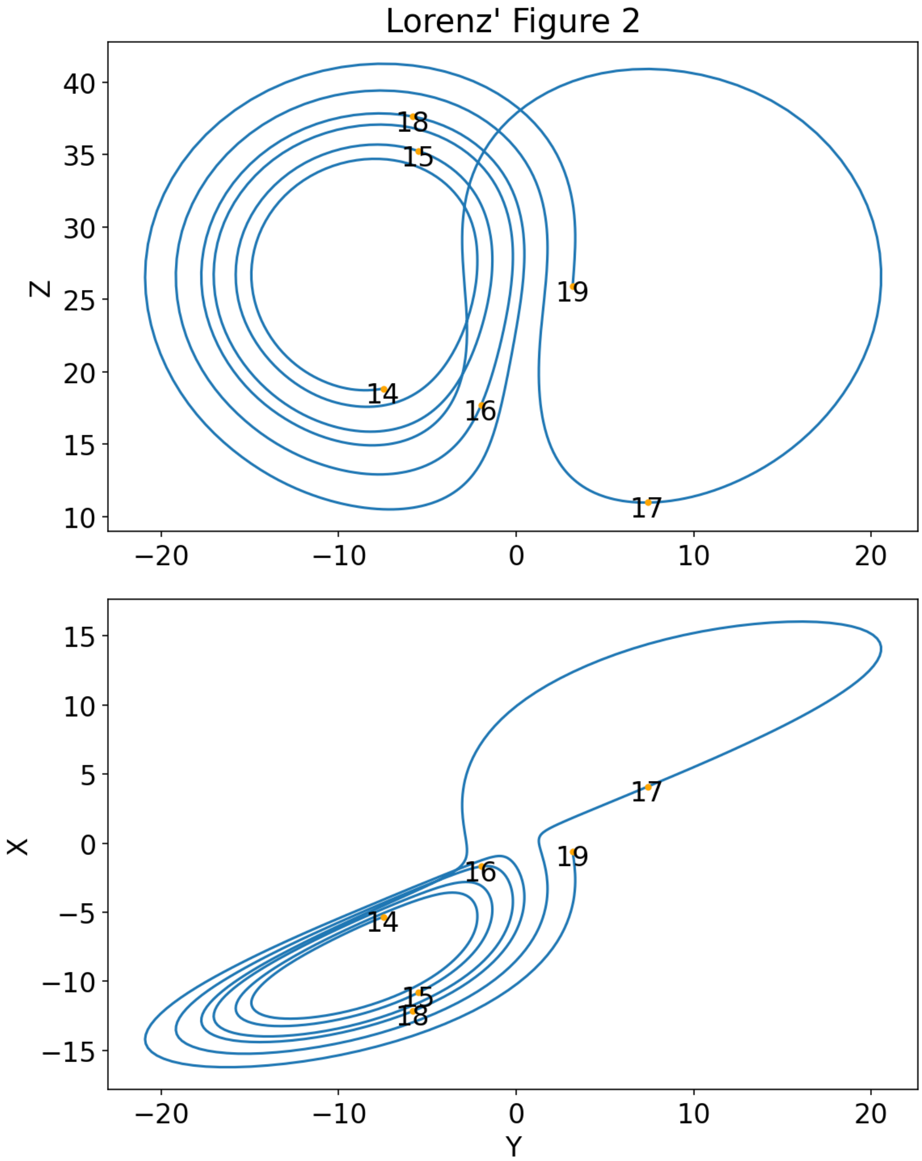

Figure 3: Reproduced Figure 2 from Lorenz' paper. Numerical solution of the convection equations. Projections on the Y-Z and X-Y planes of phase space are shown for the time interval 14-19.

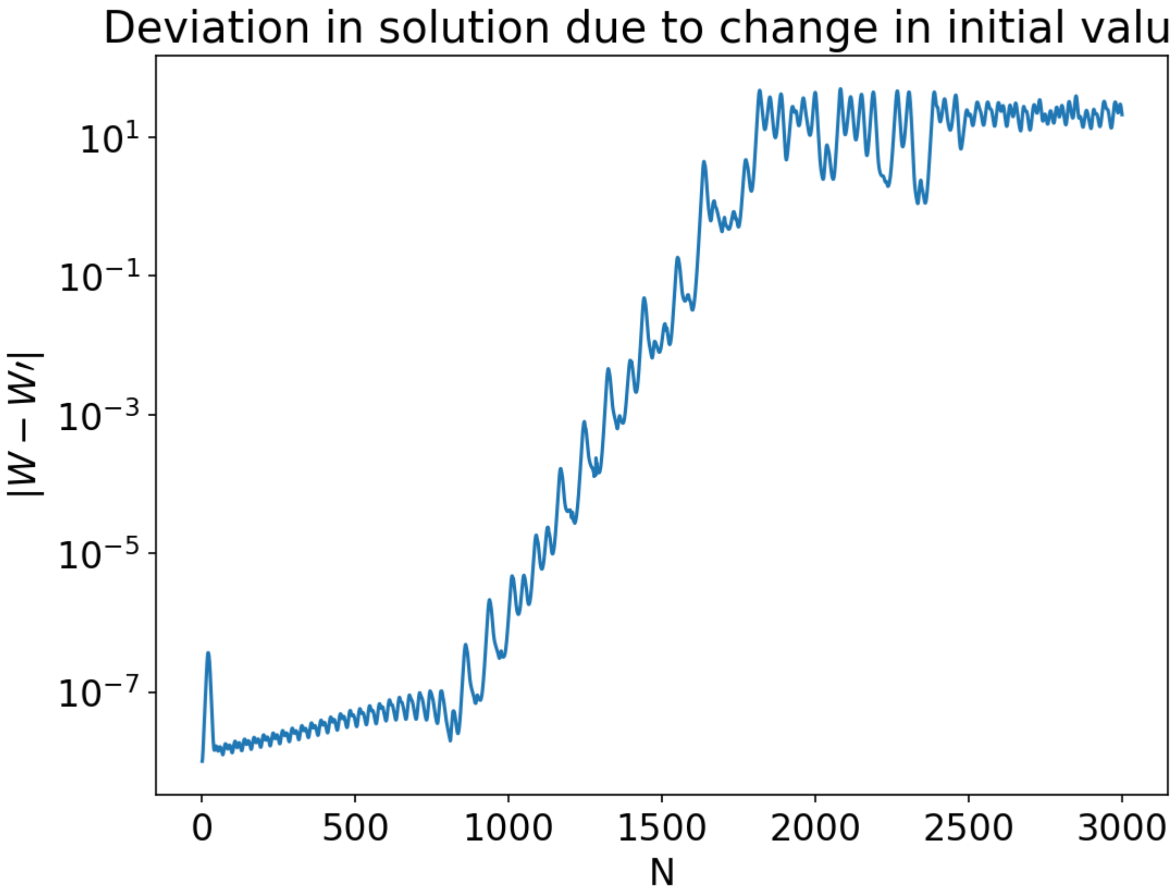

Figure 4: Distance between two solutions two the convection equations with slightly different initial conditions. Fiducial solution has initial conditions [X, Y, Z] = [0, 1, 0] and primed solution has initial conditions [X, Y, Z] = [0, 1 + 1e-8, 0]. Plotted on semilog axes (linear x-axis, log y-axis) such that the exponential growth in distance is observable.